

## Отчет по практике

### Напишем конвертер валют

Для начала напишем код:

```
import flet as ft
import requests

def get_rates():
    response = requests.get("https://api.exchangerate-api.com/v4/latest/USD")
    return response.json()["rates"]

def convert(amount, from_curr, to_curr, rates):
    if from_curr == "USD":
        return amount * rates[to_curr]
    elif to_curr == "USD":
        return amount / rates[from_curr]
    else:
        return amount * rates[to_curr] / rates[from_curr]

def main(page: ft.Page):
    rates = get_rates()
    currencies = ["USD", "RUB", "EUR", "CNY", "GBP"]

    from_dropdown = ft.Dropdown(label="From", options=[ft.dropdown.Option(c) for c in currencies], value="USD")
    to_dropdown = ft.Dropdown(label="To", options=[ft.dropdown.Option(c) for c in currencies], value="RUB")
    amount_field = ft.TextField(label="Amount", value="100")
    result_text = ft.Text("9000 RUB")

    rates_text = ft.Text("\n".join([f"1 USD = {rates[c]:.2f} {c}" for c in currencies if c != "USD"]))

    def swap_currencies(e):
        temp = from_dropdown.value
        from_dropdown.value = to_dropdown.value
        to_dropdown.value = temp
        update_result(e)

    def refresh_rates(e):
        nonlocal rates
        rates = get_rates()
        rates_text.value = "\n".join([f"1 USD = {rates[c]:.2f} {c}" for c in currencies if c != "USD"])
    
```

```
update_result(e)
```

```
swap_button = ft.ElevatedButton("⇄", on_click=swap_currencies)  
refresh_button = ft.ElevatedButton("ОБНОВИТЬ КУРС НА  
ГРИНЛАНДИЮ!!!!", on_click=refresh_rates)
```

```
def update_result(e):  
    try:  
        amt = float(amount_field.value)  
        res = convert(amt, from_dropdown.value, to_dropdown.value, rates)  
        result_text.value = f"{res:.2f} {to_dropdown.value}"  
    except:  
        result_text.value = "Invalid input"  
    page.update()
```

```
from_dropdown.on_change = update_result  
to_dropdown.on_change = update_result  
amount_field.on_change = update_result
```

```
page.add(ft.Row([from_dropdown, swap_button, to_dropdown]), amount_field,  
result_text, refresh_button, ft.Text("Current Rates:"), rates_text)
```

```
ft.app(target=main)
```

Теперь посмотрим интерфейс:

